Robotic Process Automation (RPA) Implementation Case Studies in Accounting: A Beginning to End Perspective

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ABSTRACT

Robotic Process Automation (RPA) possesses several features that may give rise to similar but also new adoption effects (both benefits and challenges) relative to other IT implementations. Our study adds to the accounting and emerging technologies literature by examining the beginning to end process through which RPA is implemented in the accounting function. We adopt an exploratory and case-centered research methodology based on a mature RPA implementation in an accounting setting. We further corroborate it with interviews with subject matter experts and compare it with another case study of an early-stage RPA implementation. We identify five themes related to RPA adoption in accounting functions: workforce, IT governance, privacy and security, system sustainability, and the measurement of RPA success. We also present key takeaways that may support effective RPA adoption in accounting. Our study provides both academics and practitioners with insights about the effects of RPA implementation.

Keywords:
Robotic Process Automation, Accounting, Workforce, IT Governance, Data Security and Privacy

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INTRODUCTION

Robotic Process Automation (RPA) refers to an automation technology that can enable a software program to emulate human actions while interacting with computer applications to accomplish required tasks (Willcocks, Lacity, and Craig 2015; Lacity and Willcocks 2017). RPA was introduced in the early 2010s, and it has gained wide popularity in business process automation since then (Willcocks et al. 2015). The McKinsey Global Institute estimates that more than 69 percent of data processing and 64 percent of data-collection activities could feasibly be automated by RPA (McKinsey 2019). In the accounting domain, RPA is being used primarily in tax services, followed by assurance services (Cooper, Holderness, Sorensen, and Wood 2019). During the COVID-19 pandemic, accounting firms have been increasingly adopting RPA to perform the jobs that have been eliminated because of the economic downturn and the cash squeeze at companies trying to cut expenses (Cohn 2020).

Although RPA is a promising service automation technology, a study by E&Y reveals that 30% to 50% of initial RPA projects fail.[1] Given the increased popularity of RPA in accounting, it is of fundamental importance to obtain a more comprehensive view of RPA adoption to better understand its impact. Emerging research in the area focuses on the use of RPA to automate accounting processes and accountant's skills that are needed to collaborate with RPA (Cohen, Rozario, and Zhang 2019; Cooper et al. 2019; Kokina, Gilleran, Blanchette, and Stoddard 2021). We conjecture that RPA possesses features that could amplify existing IT adoption issues and introduce new ones to the accounting functions, all of which deserve particular attention. For example, the purported benefit concerning RPA's ease of implementation often leads firms to

[1] See: https://www.ey.com/en_us/consulting/five-design-principles-to-help-build-confidence-in-rpa-implement#:~:text=Despite%20widespread%20global%20adoption%20of,help%20organizations%20overcome%20this%20challenge.

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adopt RPA to generate a quick Return on Investment (ROI). This short-term strategy possibly threatens the long-term goal of implementing a sustainable technology to improve accounting quality. As a result, conducting an in-depth case study from the beginning to the end of an RPA implementation, to understand the process through which RPA is implemented, is a necessary step towards facilitating an effective adoption of RPA in accounting.

We adopt an exploratory and case-centered research methodology to explore the implications of RPA adoption in accounting (Yin 2003; Miles and Huberman 1994; Roller and Lavrakas 2015). In particular, we conduct a case study on a mature project of RPA implementation in the accounting functions from Firm A, one of Denmark's largest telecommunications providers.[2] We corroborate it with interviews of subject matter experts who have designed, implemented, and managed the RPA projects at Firm A. We further compare it with another case study of an early-stage RPA implementation in an auditing setting from firm B, one of United States top 50 public accounting firms. We identify five themes depicting the relationship between RPA and an organization, namely workforce, IT governance, data privacy and security, system sustainability, and measures of RPA success. We link the themes to lessons learned from the case study and present key takeaways that may help with effective RPA adoption in accounting.

First, we find that RPA disproportionally affects accountants with low-level versus high-level skills and that accountants initially have negative attitudes toward RPA. Second, for IT governance, we find that a Center of Excellence (COE) can become a bottleneck in slowing down bot development and implementation cycle. Furthermore, more consideration is needed concerning the auditability and internal controls of the RPA bots. Third, while we find that the confidentiality of information is considered during RPA bot development, compliance activities

[2] The authors have been granted IRB approval from their institutions for this study.

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for securing the privacy and security of the data are reactive. Fourth, we observe that although RPA can lead to short-term operational efficiency improvements, it can also create hurdles for long-term change and innovation if long-term objectives are not considered. Finally, we find that in measuring RPA success, quantitative measures, like savings in FTEs (full-time employees), and qualitative metrics, such as employee satisfaction and process quality improvement, are both considered. Based on this evidence, we propose actionable steps, including the training of employees, effective communication, and improved IT governance, that may potentially increase the likelihood of effective RPA adoption.

Our study has important academic and practical contributions. First, we add to the emerging literature on the effects of RPA in accounting by conducting an in-depth exploration of the adoption of this technology. By studying the process of RPA adoption in two distinct accounting functions, i.e., in different firms, we can obtain a more comprehensive understanding of its benefits and challenges. Prior research provides valuable insights on the development of RPA applications for accounting, the skills that accountants should possess to working alongside the RPA, and accountants' use and perceptions of RPA (Cohen et al. 2019; Cooper et al. 2019; Huang and Vasarhelyi 2019; Kokina and Blanchette 2019; Kokina et al. 2021; Zhang, Thomas, and Vasarhelyi 2021; Cooper, Holderness, Sorensen, and Wood 2021; Eulerich, Pawlowski, Waddoups, and Wood 2021). We add to prior research by examining the process through which RPA is implemented, from the start of the RPA implementation to its completion, rather than at fragmented points in time**.**

Our study corroborates evidence from prior research concerning the effects of RPA implementation on workforce, IT governance, and measures of RPA success while establishing several new perspectives to these themes and introducing new ones as well. Specifically, for

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workforce we find that RPA is not likely to result in job losses for highly skilled accountants. For IT governance, we find that some internal controls to reduce the risk of error of RPA bots are considered as part of the process of RPA implementation. Finally, for measures of RPA success we find that both qualitative and quantitative measures are important. Importantly, our study introduces new perspectives concerning RPA adoption. First, for workforce, we find that RPA impacts business process outsourcing by bringing back the process in-house, through automation with RPA. Second, for IT governance we find that preventive internal controls activities to hinder the errors of RPA bots are largely missing and rather, data processing issues with RPA bots are handled on an ad-hoc basis (reactive). Similarly, we find that although the COE was established with the objective to facilitate the RPA implementation process, it was a bottleneck to effective RPA bot deployment. Furthermore, we add new insights to RPA implementation related to the security and privacy of the data that is processed by RPA bots and the sustainability of RPA solutions, neither of which have been documented in the prior literature.

Next, we draw on the lessons learned from the RPA implementations and propose actions to help optimize the use of RPA in accounting. While it is important to understand the effects of this technology on accounting, it is equally important to learn from prior RPA adoptions to draw knowledge about what worked and did not work. This study draws on the lessons learned from the case studies on RPA implementations to develop actionable steps that firms can take to facilitate the use of RPA in accounting. To our knowledge our study is among the first to propose actions that can be implemented to facilitate the adoption of RPA.

ROBOTIC PROCESS AUTOMATION IN ACCOUNTING

RPA has been widely adopted in a diverse set of business functions, such as accounting, human resources, finance, compliance, supply chain management, marketing, procurement, and

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IT. Many processes in these functions are standardized (McKinsey 2019; Protiviti 2019; Kokina et al. 2019). In terms of industry, financial services and technology/media/telecommunications companies are the farthest ahead in RPA adoption, followed by healthcare, retail, and manufacturing and distribution organizations (Protiviti 2019). In 2020, the world's leading research and advisory company, Gartner, expects that the worldwide RPA market will grow at double-digit rates through 2024, and that 90% of large organizations globally will have adopted RPA in some form by 2022 (Gartner 2020).

Recent years have seen a surge of RPA research in accounting. Table 1 summarizes the most relevant research and how this paper adds to the literature. Cooper et al. (2019) conduct interviews with RPA leaders at Big 4 accounting firms to investigate the implementation of RPA in public accounting, and they mainly find that in public accounting, tax and advisory services are leading RPA adoption, followed by assurance services. In other interviews with RPA adopters, Kokina and Blanchette (2019) explore emerging themes surrounding bot implementation for accounting, and they find that "organizations engage in standardization and optimization of processes, develop scorecard-like tools to rank tasks, adjust governance structures to include digital employees, and redefine internal controls." Moreover, they find that organizations only benefit from automating processes that are structured, repeated, rules-based, and with digital inputs. Kokina et al. (2021) explore how RPA is changing the work of accountants, they identify the roles that accountants will play in their organizations' digital transformations and categorize the skills and competencies that accountants will need to develop to successfully work alongside their digital colleagues. Cooper et al. (2021) examine how the adoption and use of RPA is affecting the perceived work experience of firm leaders and lower-level employees at Big 4 accounting firms; they find that employees and leaders both believe that "RPA is positively changing the work

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employees perform and improving employee career prospects", whereas lower-level employees report no such improvements. Other research in RPA and accounting proposes RPA frameworks and develops RPA bots to guide auditors' implementation of the technology (Moffitt, Rozario, and Vasarhelyi 2018; Huang and Vasarhelyi 2019; Zhang 2019). Moffitt et al. (2018) provide a framework to implement RPA in auditing. Huang and Vasarhelyi (2019) and Zhang (2019) provide frameworks to implement RPA and IPA (intelligent process automation) in auditing; they validate these frameworks by illustrating their application in selected auditing tasks, such as the confirmation process. More recently, Eurelich et al. (2021) propose and validate a framework to guide auditors in identifying tasks that would benefit from RPA. They find that most auditors, approximately 90%, would use it, providing evidence of its usability.

While extant literature provides important knowledge into effects of RPA adoption in accounting, we add to prior research by studying it from a more comprehensive standpoint. We observe the implementation of RPA from the beginning stages of the project to its completion, rather than at fragmented points in time, and we corroborate our observation with feedback from subject matter experts involved in the project. We further compare the case study with one from a more recent RPA implementation. Doing so allows us to obtain an in-depth understanding about the process through which RPA is implemented, which is a necessary step towards facilitating its use in accounting.

[Insert Table 1 Here]

RESEARCH METHODOLOGY

Firm A Case Study - Background

Firm A is one of Denmark's largest telecommunications providers. The company provides private landlines between houses in Copenhagen. Since the fall of 2016, the four largest business

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units of Firm A, one of which included the Finance unit, started exploring potentially using RPA as an alternative option to business process outsourcing. In this paper, we focus on the firm's adoption of RPA in accounting-related functions in the Finance unit. Firm A was selected because one of the authors was not only familiar with the various stakeholders involved in the RPA implementation and as such was able to acquire the necessary approvals for us to interview them, but also played the role of an observant of the RPA project implementation. Moreover, as the project was completed several years ago, it was possible to discuss the RPA implementation from the beginning to the end, as well as obtain feedback related to post-implementation implications. That enabled us to conduct a more thorough and in-depth analysis of the process and its results.

Data Collection and Analysis Methods

We gathered data on the implementation of RPA at Firm A through observation, review of internal documents, and interviews. One of the authors was an observant of the RPA project and was involved throughout the implementation of RPA at Firm A.[3] To complement the observation, Firm A also provided us with internal documents of the RPA development process, including materials related to RPA use cases identification, RPA methodology, and operating models. The case study is further corroborated by six in-depth and semi-structured interviews with employees who were deeply involved with Firm A's RPA development. We developed interview questions based on phenomena observed from Firm A's RPA implementation and discussions with RPA experts. We further incorporated insights from professional and academic articles that explore the effects of RPA on the business world in general and accounting in specific. Since the objectives of the interviews were to obtain an understanding of the process of RPA adoption, we focus on questions related to the following five areas:

[3] The researcher who was an observant of the RPA project was also a consultant working for one of the Big 4 public accounting firms in Denmark.

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1) participants' experience in implementing RPA
2) the operating models to create and deploy bots,
3) the roles, responsibilities and interactions among RPA developers, business process owners and management,
4) considerations of internal controls, and
5) the effect of RPA on the workforce

The interview questions were used to solidify our observation of Firm A's RPA implementation and were of probing nature. We follow the known sponsor approach (Patton 2014; Asatiani, Kämäräinen, and Penttinen 2019) to get in touch with interview participants. Specifically, our liaison/observant from Firm A helped us identify interview subject matter experts for our study and then introduced us to them. All six interviewees played key and diverse roles in the firm's RPA development. The six interviewees were selected as they represented the various roles involved in the RPA project. Each of their roles is described in Table 2. Interviewee 1 was a factory manager but had also filled the role of a process consultant before getting promoted. Interviewee 2 was a robot developer who led the development group in Firm A's finance robot factory. Interviewee 3 was also a robot developer by role. However, at the COE, he also had the additional responsibility of robot development architecture and best practices for the entire RPA initiative. Interviewee 4 was the lead engagement consultant from a Big Four company assisting Firm A in implementing the RPA initiative. His insights should be seen as broader and put the insights from the interviewees from Firm A into a larger automation perspective. Interviewee 5 played the roles of subject matter expert (SME) and process consultant, and as such was able to provide in-depth knowledge of how RPA is perceived when implemented. Lastly, Interviewee 6 was an operational consultant. She monitored the robots in production and was responsible for reporting robot key

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performance indicators (KPIs) to the upper management in Firm A. These six interviewees were selected to cover each role in Firm A's RPA implementation setup. Hence, interviewing all the roles allowed us to gain perspective about RPA from every angle.

Table 2 presents the roles and responsibilities of the interviewees. The interviews were conducted via online conference calls and were recorded, each of which was of approximately one-hour duration. During each interview, one author led the conversation and the other authors added follow-up questions and took notes. All the recorded interviews were transcribed and cross-validated by the authors. To identify the implications of RPA on accounting based on observation, the internal documents obtained, and interviews, we follow prior literature and adopt a narrative analysis approach which documents and organizes collected data into concepts to identify patterns and connections (Richardson 1995; Riessman 2002, 2008; Arthur and Owen 2019).

[Insert Table 2 Here]

RESULTS

Firm A's Approach to RPA

From our observations and review of internal documents, we note that Firm A breaks down RPA implementation into ten sequential steps depicted in Figure 1.

[Insert Figure 1 Here]

RPA project team members were involved in several roles. The responsibilities for each role, based on our review of documents and observation of the process, are described as follows:

- The factory manager has the overall responsibility for the RPA initiative ranging from strategy execution, hiring, developing training programs, reporting, monitoring, and maintaining processes to prioritizing which processes to automate.

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- The process consultant is responsible for setting up workshops, mapping processes, assessing processes subject to automation, building business use cases, and requesting user accounts.
- The robot developer is responsible for assessing the process maturity, developing the process in close collaboration with the SME and maintaining the processes.
- The operation consultant is responsible for establishing robotics environments. When the daily operation is running, the operation consultant will monitor the robots, release production-ready processes, and manage the robots. The operation consultant also needs to optimize the production in terms of IT policies and to help with user accounts, access to test data, and test environments.
- The business process SME is vital to the success of the project since they hold the knowledge about business' processes. Robot developers consult SME to gain clarification of the business process and ensure high-quality development progress. The SME will also perform user acceptance testing and gather test cases/data.

On average, Firm A spends one week for process maturity assessment; one to two weeks for process mapping; four to five weeks for process refining, development, and testing, and one to two weeks for process hyper-care and process release. Before the process maturity assessment starts, an overview of the process is created to form a high-level understanding of the where to identify processes to be automated. In this step, processes are listed in different business areas. An example of the process overview step is provided in Figure A1 in the Appendix.

After the process overview is created, leaders of identified processes fill out a process maturity assessment template. The templates are then consolidated, and a score is assigned to business processes based on their maturity, i.e., a well-established process, and potential to be

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automated. The most mature processes are subsequently vetted in a workshop and many processes are discarded during this phase due to poor data quality, ambiguous rules, or high complexity. An example of the process maturity assessment template is provided in Figure A2 in the Appendix.

During process mapping, the business process subject matter expert and process consultant document the process in detail into a file named process definition document (PDD). An example of the PDD is provided in Figure A3 in the Appendix. In this step, the final maturity test of the process is conducted. If the process is concluded as not mature enough, it is discarded at that point.

In process refining, development, and process testing phases, the robot developer starts coding the bots following the PDD. When there are questions regarding the PDD, the robot developer asks the SME. When changes occur to the process, PDD needs to be updated by the process consultant. The SME is also responsible for finding test cases or create test data as close to productions data as possible. Once the technical test is passed, the SME verifies if the process is accomplishing its goals and objectives.

During process hyper-care and process release, the release manager deploys the RPA automated process to production. Small batches of real cases are carefully monitored by the release manager and approved by the SME to secure the appropriate handling of errors (hyper-care). When errors occur, the Robo-Builder must fix the bug. Once the SME has accepted the RPA automated process, the release manager makes the final release of the RPA bots.

Once the RPA-based process has been released into production, the factory manager and process consultant host a formal handover to each business department. While these phases are not part of the bot development itself, they present the greatest need for management attention to realize the promised benefits of automation in general. Management needs to measure benefits

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realized within a given time frame, otherwise the automation journey will not provide the promising ROI as expected.

Identified Themes Related to RPA Adoption

In the previous section, we describe the firm's approach to RPA implementation. We now discuss the five themes relating to RPA adoption in accounting: workforce, IT governance, privacy and security, system sustainability, and the measurement of RPA success. The themes emerged from our observation, review of internal documents, and interviews with key players of Firm A's RPA adoption.

Workforce

Although it is not new that automation technologies can affect workforce, RPA may amplify this concern due to its feature of ease of implementation and scaling and the fact that it is being applied to white-collar work (Asatiani et al. 2019). The initial evidence from our case study shows that RPA disproportionally impacts accountants with low skills who usually perform low value-added and repetitive work, which is consistent with the findings from the prior literature (Sandler 2016).

Interviewee 1 described that RPA is not typically implemented with the purpose to lay off employees:

> *"I don't think I've ever seen a large-scale RPA initiative being the sole purpose of a lot of people being laid off. Some will lose their jobs. Others will shift to other places. I don't think necessarily that the ethics is around people getting laid off, because in reality, that's the nature of business. But I think the issue around sort of the societal impact is who is being laid off, because it's not so much, you know, the people who are currently, let's say,*

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the weakest in the organization, the lowest hierarchy, lowest education, most mundane jobs."

RPA has not replaced accountants in the headquarter of the firm but has resulted in layoffs of about 100 FTEs that perform outsourced accounting work in India. There were 83 FTEs in the Finance Unit of Firm A's headquarters (hereafter, Firm A Finance Unit). 25 of the FTEs were in purchase to pay (PTP) function, 42 were in the order-to-cash (OTC) function, and 16 were in record-to-report (RTR), i.e., the financial reporting function. According to the firm's internal assessment, the PTP function is the most prone to RPA, followed by RTR and OTC functions. PTP processes that have been automated include accounts payable and payment processing. In RTR, such processes are financial reconciliation, accounts closing, and consolidations. In OTC, processes that are automated with RPA are billing, bank reconciliations, and order entries. As an example of applying RPA to the order entry process: a bot sends e-mails to customers with an offer to upgrade a current product or buy an add-on product. Customers accept the offer by clicking a link in the email and then, an order is automatically sent to the bot's email. Once the bot receives the email, it will log the information, such as product description, price of the product and, date of the order, into the company's ERP system and place an order as per information from the email.

Among the 83 FTEs, 56 were doing tasks where RPA could be applied: 29 from OTC, 12 from PTP, and 5 from RTR. The Finance Unit in Firm A aimed to achieve 5-10% FTE savings from RPA adoption. Besides cost savings in FTEs, the firm also expected qualitative benefits from using RPA, especially better customer experience, accurate and more timely accounting data, and better documentation of business processes. Prior to RPA adoption, the firm had estimated up to 15% in savings of its Full Time Employees (equivalent to 82 FTE) at its Denmark headquarters across all business units, including the finance business unit, from automating repetitive and

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mundane tasks. According to our interviewees, however, after almost three years of RPA adoption there is no evidence that RPA adoption has led to net savings in FTEs in the headquarter. For example, Interviewee 4 stated:

> *"In finance, we have busy peaks, in the beginning and the end of the month we are really busy. So what we are doing, we are trying to remove the peaks, from the employees, because they are always very, very, busy. With the robot we are trying to make them have more normal work hours. We didn't actually let people go."*

Similarly, Interviewee 1 expressed:

> *"It's not about how we can cut the cost of our FTEs, but how can focus more on speeding up the process and making people save their time, but not cost savings, because people have to refocus their time."*

Interestingly, the implementation of RPA at Firm A led to the insourcing of tasks that were previously outsourced to India. Participants expressed that the work of over 100 FTEs was previously outsourced to India and that subsequently after RPA implementation, the work that was outsourced was brought back to the firm. Interviewee 4 explains this issue in the following:

> *"At that time, we had, I think roughly one hundred and twenty FTEs. We outsourced to India. And part of that was doing the ethical contemplations whether we should do it. What kind of jobs? And are we actually relieving Danish employees to send it out abroad. But when push came to shove, it was a matter of financially, it was more cost efficient to send it down and that at that time we really needed to save the bucks. And when we back source it, it's going to be done in the precise same manner, same way that it was before, except without the arrows, so to speak. And it also gives them, the Danish managers, a better*

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opportunity to actually see what is happening, because today when they coordinate with India, it's not always that easy. Now they can just go with it and ask for their report."

The company decided to halt the outsourcing practice because it expected that RPA can perform the same tasks done by the employees in India. Additionally, management expected that RPA can perform the work faster, better, and cheaper while gaining control over its own processes. Interviewees also indicated that insourcing tasks increased employee morale and satisfaction.

In sum, the firm realized net FTE savings from the tasks that were outsourced to India. This is possibly because the employees in the headquarter perform tasks that require more human judgement than those outsourced to India and that RPA implementation also created new positions in the headquarters, such as RPA process managers and RPA developers. The firm has not achieved net FTE savings from the tasks that have been conducted "in-house" in its Denmark headquarters. As a result, the evidence shows that a salient concern is the *who is being laid off* rather than the concern of RPA leading to job losses because employees of different skill levels are affected disproportionately. The landscape of the global labor market also adds complexity to the question of "who is being laid off" as many companies operate in international grounds. Our finding that RPA mainly negatively affects low-skilled workers, but not high-skilled workers is consistent with the Skill-Biased Technical Change (SBTC) hypothesis proposed from labor economics research, which suggests that routine tasks that follow a defined set of rules are susceptible to automation whereas tasks demanding high-skills and cognitive abilities are favored by workplace automation (e.g., Card and DiNardo 2002; Autor, Levy, and Murnane 2003; Beer and Mulder 2020).

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Regarding employee satisfaction, at the beginning of RPA implementation, employees in the Finance Unit in Firm A expressed negative feelings towards RPA due to the fear of job losses. Interviewee 2 explained this sentiment in the following:

> *"We spent a lot of time turning this image up of RPA. Oh, they're going to steal our jobs. We spent a lot of time communicating that RPA is not some dangerous thing stealing my job, it's actually a tool to help me in my daily work."*

Interview participants also mentioned that the negative perception amongst employees that RPA will result in job losses and many layoffs has created tension between them and RPA developers. To prevent RPA developers from automating their work, some lower-level employees often withheld information about the business process to be automated. Interviewee 3 elaborates on this issue:

> *"So we had a lot of heat in our interactions. Is it fair to obstruct and try to cheat and do small, very small, minor, minor, minor offenses? Just playing a bit dirty. Informing late, complaining. Try to charm your way out of a Q&A session and stuff like that. Is that fair behavior in order for you to meet your own local goals [as an employee]? And I would say probably not. But I do understand the situation. Yes, I do. But that's that's one dilemma. One common dilemma"*

Participants also indicated that for employees whose jobs won't be entirely replaced by RPA, communicating the objective of RPA, which is that it is not going to steal their jobs but rather to improve the process and to improve their job performance, helped to reduce the tension. For the employees that felt threatened by RPA, once it was implemented, the thinking of RPA as a threat shifted to RPA as a strength since employees realized that RPA was processing repetitive

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work all day leaving employees the opportunity to perform the more challenging aspects of the work. Interviewee 6 expressed the perception of job satisfaction in the following:

> *"I mean, if you have not to carry out very boring tasks and very manual boring tasks, then I would assume that you will be happier about going to work. And I think that was what I observed happened for some of the finance people that we helped out with the automating of some of their processes. We were sitting just behind them and I could hear them talk that, Oh, how nice. And then I don't need to do this because then I can just put it to the robot. I think that was that was very nice to hear that."*

Our finding that eventually RPA leads to improved employee satisfaction among high-skilled workers in the Denmark's headquarter is consistent with the theorical framework on employee motivation and stress proposed by Heckman and Oldham (1976), which implies that technology can increase employees' autonomy in executing tasks that require their discretions and feeling of meaningfulness, and thus, work satisfaction, by taking away the mundane and routing part of the employees' work (e.g., Hackman and Oldham 1975, 1976; Karasek et al. 1998; Beer and Mulder 2020).

IT Governance

RPA brings specific challenges that are particularly relevant to IT governance (Willcocks et al. 2015 and Asatiani et al. 2019). IT governance refers to the "patterns of authority for key IT activities in business firms, including IT infrastructure, IT use, and project management" (Sambamurthy and Zmud 1999). Governance of RPA bots is fundamental to the proper use of this technology in accounting as the RPA is likely to process large amounts of financially related information. For example, assume an RPA is consolidating sales and expenses information to

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generate an income statement, if the RPA fails to include a file for the less routine expenses, such as interest expense, the income statement is likely to be incomplete and inaccurate.

RPA may impact IT governance for the following reasons. First, although the feature of RPA to emulate UI (user interface) interactions is useful in enabling automation that can hardly be delivered by traditional automation tools, it also generates concerns about the access control and management of the bots as it is hard to differentiate between human and bots apart when bots can mimic human behaviors. Second, although RPA has enabled the democratization of automation, it could also lead to issues like shadow IT and therefore, proper governance of the bots is required (Willcocks et al. 2015; Asatiani et al. 2019; Harrast 2020; Syed et al. 2020).[4] Furthermore, the decision of RPA implementation and development could be made autonomously within business units, bypassing the company's IT department (Willcocks et al. 2015; Lacity and Willcocks 2017). This could add to the concerns about the governance and risk management of RPA (Asatiani et al. 2019).

Center of Excellence (COE)

The main approach for Firm A to establish proper governance over RPA is via a Center of Excellence (COE). After the evaluation of the RPA initiative in early 2017, a COE was established at the company level, with decentralized RPA teams in each department. Interview participants indicated that a consulting firm assisted with developing a framework for RPA governance. The purpose of this framework, which is consistent with the objective of the COE, was to create transparency and a common standard for the development and implementation of robots. The goal of the COE was to provide the best practice and governance of RPA bots by owning the automation

[4] According to information systems literature (e.g., Silic and Back 2014), shadow IT refers to "all hardware, software, or any other solutions used by employees inside of the organizational ecosystem which have not received any formal IT department approval."

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platform, securing the quality of code, managing the deployment of the robots, and reporting on the benefits gains to management.

Interviewees expressed that at the beginning of RPA adoption, they expected that these compliance-related activities can improve the efficiency of the process and transparency. However, as the deployment of robots continued at the firm, the governance model enforced by the COE proved to be stringent. For example, interviewee 4 stated:

> *"We had a COE. And in the beginning, it was a very, very good to have the COE. But then, it became more of a police force, and this is not in a positive way because everything all of a sudden had to be verified and triple verified. And if we had to make a change in production, it was a lengthy process. So very quickly, within a year after we had to do a major change. The COE was removed because we couldn't work with something that was so far away from the real life because we had customers in our end that actually needed these processes. So we couldn't wait for them to have time to review it."*

Combined with the time pressure to create and deploy robots, participants eventually started circumventing the governance model. An example described by interviewee 2 was that instead of sending the robots to quality assurance for review, some employees released them directly to the development environment:

> *"All of these cases had to be completed within two days. Then we cannot wait for the COE to make a time slot to get this review, so run it on your local machine in development and make it make it happen. And people know that was not quite good, but that was what was demanded of them."*

The COE was perceived as a 'check the box' process that created bottlenecks and little improvements to the RPA implementation process. Ultimately, the dissonance between RPA

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governance activities and the need to produce robots in a short period of time resulted in shifting the responsibility of robot development and deployment back to the local business units. As stated by interviewee 1:

> *"The COE was like sort of a bottleneck. So that made some friction because the procedures. It was nice to have the procedures like lined up and transparency and all. But they were not executed in an efficient way. So that that is where it sort of began to deviate."*

Similarly, interviewee 2 expressed:

> *"So all the RPA managers in the business were saying this, we cannot deliver our business goals with having this control of this COE because everything was like waiting for the COE. So now that risk falls into the business lines, if they want to like speed ahead, then they have to take all the risk and all the technical debt."*

The shift of responsibility decreased the turnaround time of robot production from months, under the COE, to days, at the local business unit level. Interviewees indicated that the main challenge with decentralizing the monitoring of the production of the robots, however, was that there were no specific compliance activities for the process. An example often illustrated by participants was that documentation was no longer required. Instead, after the robots were programmed, developers had the ability to change their operating aspects. For example, interviewee 4 described the following:

> *"And to be frank, from our end we don't use the documentation after it's been programmed because it's much easier for a developer if I need to go in and change something, I would just go into the process and step through it and see where it's failing. So now we don't spend much time on process documentation."*

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Since such activities were not recorded, interviewee 4 stated that it would be possible for developers to potentially make unorthodox changes to the robots:

"You could bypass it if you want to. But for now, I chose to trust the developers because otherwise you would go back to the old COE and then and nothing gets done."

We interpret the finding that the firm-wide COE failed to serve its expected purposes and that it was eventually broken down into business-unit-level COE to be consistent with the institutional theory and resource dependence theory. First, the institutional theory suggests that organizations tend to adopt standard operating procedures that meet legislative requirements but do not improve the organization's internal improvements (Zucker 1986; Weerakkody, Dwivedi, and Irani 2009). For example, prior research has found that government organizations implement management accounting systems to satisfy legislative requirements but make little use of the systems to improve accountability or decision making (e.g., Cavalluzzo and Ittner 2004). In Firm A's case, the setup of the firm-wide COE for RPA is to meet the compliance or legislative requirements of IT governance, but this centralized and compliance-oriented function failed to meet the needs of individual business units and eventually became a "check the box" bottleneck.

Second, replacing a firm-wide COE with localized COE is consistent with implications of the resource dependence theory, which asserts that management can enact to reduce recourse interdependence and uncertainty via means such as mergers/vertical integration (e.g., Pfeffer and Salancik 1978; Hillman, Withers, and Collins 2009). In the case of Firm A, with the existence of the firm-wide COE, the bot deployment at individual business units needs to rely on the performance efficiency and effectiveness of this firm-wide COE, limiting the control of the business unit management over the bot implementation. Thus, replacing the firm-wide COE with

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localized COE at the business unit level can enable higher levels of resource control of local management, enabling more flexible and efficient bot deployment.

Auditability and Internal Controls

In terms of the auditability of the output produced by the bots, a weekly report that outlines the activities that the robot executed, and any exceptions is generated. The report is subsequently reviewed by the team lead to evaluate exceptions. This internal control activity, of detective nature, checks the output that is generated by the RPA and to some extent, the appropriateness of its development and production. However, there were no preventive internal control activities identified other than segregation of duties relating to the authorized access by IT users such as the developer and deployment manager.

Based on our observation and interviews, it appears that the auditing of the bot development/production and of the RPA output has not been extensively considered. Interviewee 5 stated that:

"Internal audit was not present in our discussion and in our governance structure."

Similarly, interviewee 6 expressed:

> *"Management knew about possible risks. We talked a lot with Finance about the bots that should kind of reimburse customers with real money so it was not that it had no risk, especially that [bot], you needed the segregation of duties there to be quite significant, not just kind of the normal, ok, both the developer and the deployment manager do not have access. So it's a bit odd that we weren't more surrounded or at least consulted with the internal audit. They also did not audit our reporting."*

Interviewees indicated that RPA scripts produce a lot of structured data and that such data can be used to continuously monitor the flow of the business process. In considering the

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auditability of the development and production of robots, interviewees stated that the focus should be on the activities that the robot is executing. An example described by interviewee 1 is that if the robot is programmed to make updates to the human resources (HR) system for terminations, it will not lead to severe and potentially detrimental outcomes to the firm. However, if it is processing transactional-related work, i.e., accounting or finance related, then its design and execution should be audited and audited frequently:

> *"The whole question around auditing bots for me is it's really should be focused on. What is the severity of the implications of what the bot is doing? Because sometimes the bot is doing stuff that it doesn't really matter, like for HR. If it's doing it wrong, then somebody is to know, somebody is just becoming a bottleneck of some silly automation. And if the bot is doing transactional work, if it's doing finance impacting work, I think it should be audited in a much more frequent a perspective because the audit itself should be much easier."*

Interviewees were aware of the division of roles among IT personnel, i.e., segregation of duties, to ensure that no employees had unauthorized access to applications. An example described by interviewee 3 was that developers had to request access approval to application owners before they could begin working in an application. The level of access depended upon the type of activities to be automated by RPA, but in general access was restricted to the particular activities that needed to be automated:

> *"Every time that we work with a certain application, we have to talk to the application owner to make sure that they will approve us working into the system. And it's been even more important now because we, the whole company has been segregated into two different companies. We have to make agreements with each side and write a contract with them if*

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we want to work into their applications. So we still need approval from them in regards to the applications. We request all application access through our HDSM system and then management and group security will look into our request and then they either will give accept it or they will decline with an explanation why."

Finally, interviewee 1 expressed that the internal controls designed to prevent developers and end users from maliciously programming the robots from stealing cash for example were crucial and that internal controls might serve as a safeguard to prevent human error or fraud.

"You can say. If it was designed to take some spreadsheet and punch in some numbers based on the values in the spreadsheet, then you could just give it malicious like data and say, I want to discount my model like very much. So that's another side of it, so we couldn't, like take out all the fraud scenarios."

To summarize, it appears that there are some internal controls that are implemented to reduce the risk of error of the RPA bot, including segregation of duties at the IT access level, and internal controls to check the output of the RPA. However, while there is awareness regarding the auditing of the RPA bots themselves, audits of the bots are not performed. Moreover, preventive internal control activities to reduce the error of RPA bots that process financial information are not in place suggesting that governance of RPA can be improved.

Data Privacy and Security

According to Protiviti (2019), data security concern is among the top obstacles of RPA adoption. RPA can operate on different applications throughout an organization (Willcocks et al. 2015; Lacity and Willcocks 2017). This interoperability feature of RPA can enable large-scale automation across different applications, but it also spurs concerns about maintaining the data privacy and security since RPA makes it easier to move data across applications compared to the

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manual process. Besides, the bots can act like human workers, and they usually have the same credentials as human workers. If RPA is adopted to automate tasks involved with sensitive data, the risk of the bots mishandling (either due to intentional fraud or unintentional errors) the data should be taken into consideration.

Employees were aware that RPA is a data-driven technology that has implications on data privacy and security. Interviewee 3 expressed that the development and production of RPA robots had to comply with GDPR (General Data Protection Regulation) standards.

> *"We also had to record a lot of processes in regards to GDPR. So that we are not breaking the law on that one. We have had a data breach. When it happened, we had a department only working with GDPR. And then they have given us some governance, which we have worked after. So then you make sure that does not happen again."*

Interviewee 3 mentioned that Firm A experienced a data breach incident because of programming errors in RPA robots:

> *"A bot was incorrectly programmed and used the wrong customer information so the customer saw other customers' information, which violates the requirements of GDPR. We found out because the customer contacted the call center and then the news went to the process owner and bot developer. The issue was remediated by reviewing and modifying the code of the bot and it was not captured before likely because of poor communication. The developer just followed instructions provided by the process owner and there was not enough pilot testing."*

Interviewee 3 and interviewee 4 stated that after the data breach incident, compliance activities such as the backing up of data every six months and HR (human resources) training on

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authorized usage of data were in place to minimize the risk of intentional or unintentional data breaches. Interviewee 3 expressed:

> *"Also about locking the data, how long can we lock data. We got a lot of governance on that part. Every six months, we have to take a course which we must pass from H.R. in regards to GDPR so that we are up to date on what we can and cannot do."*

Similarly, interviewee 4 stated:

> *"Every half year we delete all the information that was created half a year ago. So we only save the last six month worth of data. Because otherwise, it would be too constraining for us that we need to go in and delete the logs every time."*

While there was awareness on the importance of maintaining data privacy and security, a data breach occurred, and compliance procedures were implemented reactively following the incident. This highlights the necessity to carefully consider the effects of RPA on confidential information.

System Sustainability

Interviewees acknowledged that RPA led to short-term operational efficiencies in business processes. As an example, the firm had several ERP systems that were not integrated. Though the integration of these systems could be accomplished with traditional IT, interviewees explained that this could be extremely costly and time consuming, whereas with RPA, implementation is cheaper and less time consuming. Interviewees also mentioned that RPA can be considered as a digital workforce that can perform tasks that humans don't have time to do to improve operational efficiency.

For example, interviewee 1 described how RPA is perceived to be a short-term solution:

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"You know, it's a Band-Aid for a very big open wound. So why are we using RPA? Well. Because our ERP systems are not integrated with our old mainframe systems. And in order for us to do that, it costs, you know, a couple million. It will take six to eight months, etc. So instead. Ok. Here's a quick fix. And then let's get cracking and then we realize, hey, the quick fix can also actually support many other places. "

While RPA helped to realize synergies within business processes, it created the IT challenge of updating and maintaining the bots. The common view by employees about RPA was that it is a 'quick fix' type of solution. We found that RPA bots require constant maintenance because any changes to the business process or underlying application may require the bot to be modified and that upgrading systems where the bots operate may be avoided because that may render existing bots obsolete.

Interviewee 2 expressed that building a durable robot is not easy:

"RPA is easy to use software. Yes. Comparatively, it is perhaps easier to use, but sometimes it's also the promise of how easy it is to build robots. I think it's exaggerated. I think it's harder to get a durable, maintainable robot. They have like a huge headache maintaining these robots. Because why are we going to do that. There's this robot, it has 1000 boxes, but it's sort of doing what it's supposed to be doing. but you could not change a thing. Because then the whole thing would tumble down. So this is not the right way to design a bot."

Similarly, interviewee 6 indicated that maintaining RPA bots took a lot of time:

"I think one of the things that I always realized during my time with RPA was the significant amount of work that it required to keep the robots running. I mean, I think there was quite

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a lot of maintenance in order to make sure that they were executed on time. And I think that kind of surprised me."

Since employees did not spend sufficient time in understanding how RPA technology could seamlessly be integrated with ERP systems, ERP system changes created technical inefficiencies. Thus, RPA may be more valuable when the company does not want to fundamentally change the underlying IT infrastructure in the long-term.

Interviewee 1 attributed the lack of change management and the time pressure to build robots as contributors that might have diminished the success of RPA deployment as a sustainable solution.

"But I think that it can be a major issue if you don't align the strategy of the implementation of RPA with also the change management aspects of it. I think change management could have been a driver for faster success."

Building durable and maintainable robots is a worthwhile time investment that can more closely align with the long-term strategic goals of the firm. RPA as a technology is user friendly with its minimal (but not absent) requirements of technical skills. Creating an RPA bot involves less coding than full automation, however it may also require more frequent maintenance due to its reliance on the underlying processes. Most of the existing literature focuses on the feedback obtained from interviewees, which was likely based on their early experience with RPA, potentially before any long-term system issues were considered. For instance, despite the user friendliness of RPA, keeping the bot from breaking requires a tremendous amount of time and effort to address any updates/upgrades to the system. In certain cases, the involvements of the organization's IT department would be necessary, introducing additional delays and potential complications. Consequently, our interviewees indicated that RPA would be a suitable partial

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solution to address current automation needs, however as the underlying processes or the systems used to run them change over time, the sustainability of RPA becomes questionable. To put it differently, RPA is an attractive solution to most adopting companies due to its interoperability and ability to function with legacy systems, indicating its suitability as a "quick-fix" solution (Kokina and Blanchette 2019). However, these features make RPA susceptible to changes in any of the involved systems, rendering it a short-term solution and impacting its long-term sustainability.

Measures of RPA Success

RPA success at Firm A was measured by using quantitative metrics relating to FTE savings. Interestingly, qualitative measures of employee satisfaction, customer satisfaction, compliance, and process quality improvement were just as important. By the time the project was completed, Firm A had saved about 100 FTEs for the work that they outsourced to India but did not reduce FTEs in their Denmark headquarters. The interviewees mentioned that although no FTEs were reduced in the headquarters, the firm values the qualitative improvements that RPA has brought, especially relating to customer satisfaction, employee satisfaction, and process quality improvement, such as more accurate and timely accounting data, as well as better documentation of business processes. Interviewee 5 stated that RPA improved business processes and customer satisfaction:

> *"We didn't realize the expected FTE gains, but it was important to make the process run smoother, in general was a big aspect of the implementation. And I also, I feel like we had another great case in the commercial business unit where we improved the customer experience quite a bit. We had a huge commercial benefit because we were actually able to provide the customer with better experience."*

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Interviewee 4 stated that employees were more satisfied with their jobs because the bots enabled them to focus their efforts on higher value tasks:

> *"So now they're better able to actually say, OK, actually I don't need to do this because I know that this robot can do it. So we don't need to do that. So I think right now we are on good terms with the with the team leaders and also with all the SMEs, because they also let's be frank, it's not fun doing repetitive work. You want to do something more challenging. So if the robot can remove all the repetitive work, then everybody's happy and then they can concentrate on the things that's more challenging and requires more investigation and so on."*

RPA also produces more accurate accounting information as stated by Interviewee 5:

> *"Customers paid for their subscriptions twice for I don't know how long and of course we wanted to pay them their money back. But at the end of the day, well, it was also money on our balance sheet. It was not prioritized. However, now with the robot doing that, we could just run that query to get them repaid or reimbursed."*

As illustrated, it is important to measure IT success using appropriate metrics (Willcocks and Lester 1999). Extant RPA studies primarily focus on quantitative metrics. Examples of quantitative metrics used in RPA projects are: Return on Investment (ROI) (e.g., Lacity, Willcocks, and Craig 2015), savings of Full-Time Equivalent (FTE) (e.g., Dunlap and Lacity 2017; Schmitz, Dietze, and Czarnecki 2019), overall cost-savings (e.g., Dunlap and Lacity 2017; Bruno, Johnson, and Hesley 2017; Schmitz et al. 2019), automation percentage (e.g., Schmitz et al. 2019), and time-savings (e.g., Anagnoste 2018). Besides metrics that can be quantitatively measured, we find that

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qualitative metrics of RPA success can be useful. Consequently, both quantitative and qualitative metrics should be considered in RPA projects to adopt RPA at scale.[5]

Other *Themes - The effect of RPA on trust and communication*

During our interviews, it was expressed that trust needed to be created to help employees understand why the implementation of robots might lead to synergies in the firm. During the development and production process of robots, employees appeared resistant to change and they often withheld information from RPA developers, primarily because of the fear that they would lose their jobs. Another issue related to trust is that now employees had to rely on and therefore trust the bot to do the job that they used to do. In addition, the interviewees indicated that poor communication during the bot development phase caused the previously mentioned data security breach. It is noteworthy to mention that the tension between developers and business process owners decreased once they began to effectively communicate the objective for using bots as part of their jobs. [6] The identified themes and main takeaways are summarized in Table 3.

[Insert Table 3 Here]

COMPARATIVE CASE STUDY

We compare our research findings in Firm A to a case study of a recent RPA implementation. The case study firm (hereafter, Firm B) is a public accounting firm in the United States that implemented RPA in one of its assurance practices. Two of the co-authors of this paper observed the bot design, development, and deployment phases for this firm.

While Firm B's project is more recent, it has the advantage of the involvement of two of the authors in the RPA project from its inception to the deployment phase. As a result, it was

[5] For more details, please visit: https://www.protiviti.com/US-en/insights/making-rpa-sustainable

[6] Poor communication can also happen unintentionally because employees fail to provide the details of the process that the bot developer needs to configure the bot.

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possible not only to interview but to observe the entire RPA implementation project. Moreover, since most of Firm B's clients are non-public companies, the firm faces less regulatory challenges with regards to new technology implementation than a public accounting firm that audits public companies. It is noteworthy that there exist certain differences in the fields of operation of Firm A and Firm B. Firm A is in the telecommunication industry and Firm B is in the Auditing industry. Thus, the nature of services they provide introduces variability as to how they approach implementation of RPA. Moreover, Firm A is a bigger company than Firm B and as such has more resources than Firm B. Despite these differences, Firm B is a suitable choice for comparison to Firm A since we are interested in comparing implementations of RPA to an accounting related process.

Firm B's use case consists of the applications of an RPA bot to one of the processes of auditing real-estate clients. This process can be described at a high level as follows: auditors upload working papers to a tax software, run various calculations, download a file with results generated by the software, and finally, reformat this file based on the company's requirements. Several of the challenges encountered for Firm B during the RPA development and deployment stages are similar to those identified from the Firm A's case, but not all.

Consistent with what we find in Firm A, we observed that auditors in Firm B initially showed concerns about bringing automation to their day-to-day work. However, after they were shown the purpose and benefits of RPA, such as assisting them in dealing with mundane repetitive tasks, they became more receptive of it. It is interesting though that unlike Firm A, the employees in Firm B did not express any job security concerns. Moreover, the RPA implementation in Firm B did not lead to any job losses, unlike Firm A. One of the possible explanations is that the scope

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of the RPA project in Firm B was more limited in comparison to Firm A and did not extend to the business processes that were outsourced.

Similar to Firm A, communication challenges between bot developers and auditors were also identified in Firm B: the bot had to be reconfigured and revised multiple times because the auditors initially failed to show the bot developers all the variants of the process to be automated. This issue was resolved after the auditors learned the basics of RPA and the bot developers got acquainted with the underlying audit process. During the final testing stage of the bot, an error was identified. This error was caused by the bot as the developers had not included a process variant in it. Since the error was identified before the official bot deployment, it was possible to adjust the bot in time to prevent further errors.

In addition to the above challenges identified in Firm B, we especially noticed the challenges of RPA success measurement and sustainability. In measuring the benefits of RPA, efficiency improvement was first considered in Firm B. However, since the software program that RPA interacted with required long response and processing time, the efficiency could not be significantly improved. To address this issue, the IT team of Firm B moved the bot to a remote (virtual) desktop to enable the auditors to attend to other tasks (on their local machines) while the bot completes the process on a remote desktop. Consequently, the efficiency improvement resulting from the bot deployment became of less relevance because RPA enables auditors to do more high-value tasks. This suggests that like how Firm A measured RPA success, Firm B also regarded qualitative benefits of RPA as important as the quantitative ones. An interesting finding of this study is that Firm B experienced a low adoption among its employees compared to Firm A. Two factors are the potential reasons for the low adoption of RPA at Firm B. The first factor is the strategy that Firm A followed where the use of the RPA was mandated for the whole unit by

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management, whereas Firm B left its adoption voluntary. It was in fact up to the individual audit partners to require their teams to use it. The second factor is the nature of the end-users, who are in the case of Firm B mainly auditors. The latter group are more risk averse than their counterpart from Firm A. As a result, Firm B ended up with a limited user adoption of the bot.

An unintended consequence from the involvement of the IT department of Firm B is that it played the role of the COE in Firm A (Firm B did not have an office COE). However, unlike Firm A, this pseudo-COE did not become a bottleneck, possibly due to the small size of the company and the low adoption that was discussed earlier. Moreover, Firm B was able to implement preventive and detective internal controls around the RPA bot. Plus, the IT department provides a log of activities to ensure auditability.

Notwithstanding the benefits of the RPA bot, its sustainability turned out to be a major concern. Since the bot interacted with the tax software's graphical user interface, it needed to be reconfigured or revised whenever the latter is to be updated or modified. The RPA project leader in Firm B indicated that this could potentially be a concern regarding future upgrades of the tax software or other improvements that could break the bot. Table 4 summarizes the comparison of the findings from Firm B and Firm A.

[Insert Table 4 Here]

DISCUSSION: HOW IS RPA SIMILAR OR DIFFERENT FROM TRADITIONAL IT

In this section, we discuss how the themes of RPA implementation identified from this research are similar and different from issues related to traditional IT adoption. In terms of workforce, like other automation technologies (Sandler 2016), RPA disproportionally impacts knowledge workers with low skills who usually perform low value-added and repetitive work.

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However, RPA amplifies this concern because this technology is easier to implement and scale, and it can automate tasks that could not be automated before (Asatiani et al. 2019). Like other automation technologies, although RPA can replace or reduce certain types of work, new occupations will emerge (Bughin et al. 2018; Daugherty and Wilson 2018; Frey 2020), for example, robot factory manager, RPA developer, and process consultant.

Like other automation technologies, RPA also requires proper governance. However, unlike managing large-scale IT projects such as ERP systems (e.g., Law, Chen, and Wu 2010), based on our case studies, governing RPA needs more flexibility and agility. Moreover, RPA poses unique risk to data security and privacy since it operates on different applications throughout an organization (Willcocks et al. 2015a; Lacity and Willcocks 2017). This interoperability feature of RPA can enable large-scale automation across different applications, but it also spurs concerns about maintaining the data privacy and security since RPA makes it easier to move data across applications compared to the manual process. And this concern is illustrated from the data breach incident from Firm A. Besides, the bots can act like human workers, and they usually have the same credentials as human workers. If RPA is adopted to automate tasks involved with sensitive data, the risk of the bots mishandling (either due to intentional fraud or unintentional errors) the data should be taken into consideration. While RPA could render challenges related to data privacy and security, it can also help mitigate such issues when it is used to protect data privacy and security. For example, a bot can be built to restrict unauthorized users from accessing and manipulating sensitive data; to add an extra layer of encryption to secure data usage; and to generate real-time reports about system usage.[7]

[7] https://electroneek.com/blog/business-insights/5-ways-rpa-reduces-cybersecurity-threats-in-2020/#:~:text=Elimination%20of%20unauthorized%20access&text=Software%20robots%20can%20allow%20only,attacks%20from%20outside%20the%20company.

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Although RPA is in general easier, faster, and cheaper to implement than traditional large-scale IT systems (e.g., ERP and CRM), it is also more prone to changes in business processes as such changes may render the bot obsolete and need to be reprogramed (Lacity and Willcocks 2015). Thus, RPA can hardly become a long-term solution for business process automation even if it excels at digitizing and automation in the short-term for stable processes.

Like measuring the success of most IT projects, evaluating the benefit of RPA requires both quantitative and qualitative measures. Based on our case studies, since RPA projects are by and large much cheaper than large-scale IT systems, the cost benefit consideration for RPA can be less focused on quantitative achievements. Rather, qualitative factors, especially customer/employee satisfaction and process improvement, often play an important role in determining RPA's success.

FUTURE RESEARCH DIRECTIONS

This research is qualitative and exploratory in nature, and it, therefore, sheds light on several directions for future research. The agreements and disagreements of the findings across the two case studies highlight the contextual aspect of RPA implementations and emphasize the idiosyncrasies of such endeavors. First, as more firms start to adopt RPA in their business operations and accounting functions, future research may be able to perform archival analysis to empirically examine the impact of RPA on the accounting workforce in aspects like job replacement and work satisfaction. Furthermore, since there are often ethical debates when it comes to technology's impacts on human workforce, future research can also explore the ethical implications of RPA to the accounting workforce. Third, the two case studies from this research showed that RPA can create new risks related to data security and privacy due to its interoperability feature, but at the same time, it may also decrease such risk by automating controls that can ensure

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data security. Thus, this tension leaves an avenue for future research to empirically examine the net impact of RPA adoption on the risk of data security. Fourth, while RPA is increasingly used to assist accounting work, it also not clear how RPA will impact accounting quality. On the one hand, by standardizing and automating accounting procedures, RPA could lead to better accounting quality in terms of less error and improved efficiency; on the other hand, RPA could also hamper accounting quality in the means of coding errors or malfunctions. Therefore, further research can also study the impact of RPA on accounting quality. Another venue for research is examining the driving forces for RPA adoption in accounting, especially when the company adopts a normative strategy requiring the use of the RPA bot. Finally, future studies can empirically examine some of the factors that make successful (failed) RPA projects, some of which would include the company's investment in the RPA technology, software used, and whether to contract expert bot developers as opposed to developing it in-house. Table 3 provides a summary of future research directions corresponding to the RPA adoption themes identified from our case studies.

CONCLUSION

We examine the process through which RPA is implemented in the accounting function by conducting a case study of RPA implementation from the beginning stages of the project to its completion. We corroborate it with interviews of subject matter experts who were involved in the project and further compare the case with an early-stage RPA project in an auditing setting. Based on observations, review of internal documents, and interviews, we identify five themes depicting the impact of RPA on the accounting function: workforce, IT governance, data privacy and security, system sustainability, and measures of RPA success. We draw on lessons learned from the case and present key takeaways that may increase the likelihood of effective adoption of RPA in accounting.

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Our study has important academic and practical contributions. First, we add to the emerging literature on the effects of RPA in accounting by conducting an in-depth exploration of the adoption of this technology. Studying the process of RPA adoption enables us to obtain a more comprehensive understanding of its benefits and challenges. We also extend existing literature by introducing new perspectives to the themes of workforce, IT governance, and measures of RPA and by introducing new themes – data security and privacy of the data that is processed by RPA bots and the sustainability of RPA solutions. Finally, the key takeaways provided in this paper present actionable steps that may be useful to practitioners who are adopting RPA. We believe that firms that expect to implement RPA or are currently implementing RPA in their finance/accounting function would benefit from the themes and takeaways in our paper.

Since this is a case-centered study, one limitation is that our findings may not be generalized to RPA implementations in other settings. However, we try to address this concern by complementing a mature RPA implementation with an early-stage RPA project in a diverse accounting function. Another limitation is the limited number of interviewees. Although the interviewees in this study are key players in the RPA adoption of the firm, future research can conduct analysis on a larger sample and provide more descriptive evidence or initial archival evidence as RPA becomes prevalent in accounting. Future research can also study the ethical implications of RPA in accounting settings.

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TABLES AND FIGURES

Table 1. Comparison with Relevant Research

Paper	Research Objective	Research Method	Main Findings/Conclusions
Cooper et al. (2019)	Investigates RPA software implementation in public accounting	Interviews with RPA leaders from the Big 4 accounting firms	• RPA implementation was furthest adopted by tax services, followed by advisory services, and finally assurance services. • RPA had no impact on audit fees. • Clients may want a decrease in fees. • RPA implementation was mostly driven by lower-level employees. • RPA had significant improvements in efficiency and effectiveness. • Interviewees do not anticipate workforce reduction.
Kokina and Blanchette (2019)	Exploration of the major themes related to RPA implementations in accounts and finance processes.	Interviews with individuals from different accounting firms	• Prior to RPA implementation, organizations attempt to standardize and optimize their processes, create scorecards to rank tasks for RPA suitability. • Digital employees are included in the governance structures, and internal controls are subsequently redefined. • Benefits from automations is observed when the processes included in the RPA implementation are structured, repetitive, and rule-based. • Other benefits from RPA include improved process documentation, decreased human-caused errors, better reporting quality, and improved performance measurement.
Kokina et al. (2020)	• Investigates the effects of RPA on the jobs of accounting and finance professionals • Examines the role of accountants in RPA-related digital transformation • Describes the required technical skills accountants will need for a successful collaboration with their digital counterparts (i.e., Bots).	Interviews with individuals from different accounting a RPA vendor firms and review of RPA-related documents	• The roles of accountants in RPA implementations include identifying, explaining, training, maintaining, and analyzing of RPA related projects. • New technical skills are necessary for accountants to successfully perform the above-mentioned roles.
Cooper et al. (2021)	Investigates the impact of RPA adoption and use on the perceived work experience	Interviews and surveys	• General perception of the impact of RPA on the accounting profession is positive.

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	of the Big 4 accounting firms' leaders as well as lower-level employees		• Firm leaders and lower-level employees agree that RPA modifies the employees' jobs/tasks in a positive way. • Firm leaders and lower-level employees agree that RPA improves career prospects of the firm employees. • Firm leaders believe that RPA has a positive impact on job satisfaction, unlike lower-level employees who do not agree on such impact.
Moffitt et al. (2020)	• Describes the use of RPA in the auditing profession • Envisions a future of audits empowered by RPA.	Design science research	• Proposes a conceptual framework for the implementation of RPA in auditing. • Discusses the possible implications of RPA in auditing.
Zhang (2019)	• Explores the potential benefits of Intelligent Process Automation (IPA) in audit engagements • Proposes an IPA implementation framework for audit engagements	Design science research	• Proposes a framework that uses audit workflow for IPA implementation in auditing. • Illustrates the concept of IPA in auditing by providing an IPA prototype.
Huang and Vasarhelyi (2019)	Applies RPA technology in auditing	Design science research	• Proposes an RPA framework to automate repetitive and low-judgment audit tasks. • The proposed framework has the potential to free up auditor's time to target higher-value judgment requiring tasks. • Illustrates the framework using an RPA implementation on the confirmation process.
Eulerich et al. 2021	Draws upon socio-technical systems theory to develop and validate a framework to select projects that can be automated with RPA	Design science research	• RPA implementation framework is useful to approximately 90% of respondents of which included external and internal auditors.
Current paper	Explores the effects of RPA implementation in accounting	Case study with interviews where the RPA project is observed from beginning to completion and feedback is obtained from project participants	• Complements existing literature on findings related to workforce replacement, IT governance, and measures of RPA success. • Extends current studies on findings related to RPA's impact on business process outsourcing, inefficiency of COE, sustainability, and data privacy and security.

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Table 2. Interviewee's Roles and Responsibilities

Interviewees, length of interview (min)	Role	Responsibility	Experience with RPA
Interviewee 1 (60)	Factory manager and process consultant	Identify and assess RPA use cases	3 years
Interviewee 2 (55)	Lead RPA developer	Develop RPA for the finance department	3 years
Interviewee 3 (60)	COE architect and RPA developer	Design and maintain RPA COE	2 years
Interviewee 4 (62)	Lead RPA consultant from a Big 4 accounting firm	Help Firm A apply RPA	7 years
Interviewee 5 (56)	SME and process consultant	Provide knowledge about business process and establishing robotics environments	2 years
Interviewee 6 (54)	Operational consultant	Monitor the robots in production and report robot KPIs to the upper management in Firm A	2 years

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Table 3: Themes of RPA Implementations and Takeaways for Practitioners

Theme	Issue	Description	Examples	Takeaways for Accountants/Auditors	Open research questions
Workforce	Job losses	• May occur in low-skill jobs.	• Job losses happened mostly in some low-skill jobs. High-skill workers were mainly empowered rather than replaced by bots.	• Provide training to low-skill workers: to gain new skills, be repositioned, and/or improve human-bot collaboration. • Provide training to high-skill workers: to improve human-bot collaboration.	• What is the impact of RPA on the accounting workforce aspects such as job replacement and work satisfaction? • What are the ethical implications of RPA to the accounting workforce? • What is the implication of RPA on outsourcing (especially considering the job replacement that happened in outsourced positions)?
	Employees' negative perception of RPA	• Employees may have negative attitudes towards automation, even for high-skill workers.	• Some employees in the firm refused to communicate with bot developers. But were more open to RPA once they were shown its capabilities.	• Communicate with employees: management's expectations for adopting RPA. • Demonstrate the capabilities and shortcomings of the bot.	• What is the impact of RPA on accountants' work satisfaction?
IT Governance	COE becoming a bottleneck	• Center of Excellence (COE) may not work as intended.	• COE turned out to be a bottleneck as it slowed down bot development and implementation cycle.	• Localized rather than centralized COE. • Manage time pressure to create and deploy bot.	• What are the advantages of centralized RPA governance compared to decentralized RPA governance?
	Auditability and Internal Control	• Lack of documentation, internal controls, and auditability of the bots creates hurdles for bot management and quality control.	• Lack of documentation partially contributed to the erroneous bot behavior mentioned below.	• Proper documentation. • Bot internal controls and audits. • Business process internal controls and audits related to the output generated by the bot and segregation of duties.	• What is the impact of RPA on accounting quality? • What are the components of RPA governance frameworks? • Is government regulation needed to provide RPA governance standards to

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					standardize RPA governance across the country (or globally)? • What is the level of involvement of internal audit in certifying internal controls regarding RPA? • How do external auditors get comfortable with internal controls regarding RPA? • To what extent are RPA controls more rigorous when RPA outputs are related to financial statement outputs? • Do these controls fall under traditional ICFR and SOX 404 compliance and reporting?
Data Security and Privacy	Fraud and errors	• May occur due to poor design, management, development, deployment, and maintenance of the bots.	• A bot was mistakenly programmed to extract confidential information of one customer and present it to another.	• Early involvement of IT staff and business process owner(s). • Ensure RPA bot meets data security and privacy standards, such as GDPR. • Ensure pre-production quality assurance of the bot.	• What is the impact of RPA adoption on the risk of data privacy and security?
System Sustainability	RPA sustainability	• RPA can create hurdles for long-term change and innovation if long-term objectives are not considered.	• Upgrading systems where the bots operate is avoided because that may render existing bots obsolete.	• Aim for long-term process redesign in addition to RPA implementations.	• What is the long-term effect of RPA on systems sustainability?
Measures of RPA Success	Measure of success	• Solely focusing on quantifiable RPA success while	• The firm has not realized significant reduction in full time employees at their	• Use qualitative metrics: employee satisfaction, customer	• Besides time and cost savings, what are the most suitable measures of RPA success?

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		ignoring qualitative benefits may prevent RPA adoption at scale.	headquarter. However, it realizes RPA's benefits in improving employee satisfaction, customer satisfaction, compliance, and process quality.	satisfaction, compliance, process quality improvement. • Use quantitative metrics: ROI, FTE savings, overall cost-saving, automation percentage, and timesaving	• Empirical analysis of the impact of the firm's investment (large scale vs small scale, software used, external experts vs. in-house development) on the success of an RPA implementation.
Other – Trust and Communication	Communication barriers affect bot development	• May occur between bot developers and employees and can hamper effective bot development.	• Ineffective communication between employees and bot developers caused re-configuration of some bots.	• Designate some business process employees to become RPA developers. • Include RPA developers into day-to-day business activities.	• How would the type of adoption (normative vs. voluntary) impact the level of utilization of the bot?

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Table 4. Findings Comparison of Two Case Studies

Theme	Issue	Similar findings (Firm A / Firm B)	Details
Workforce	Job losses	No	No job replacement happened yet: 1) Auditors who used the bots were empowered by them, 2) there was limited adoption of the bot (adoption was not mandated), and 3) RPA was implemented for a single process.
	Employees' negative perception of RPA	Yes	Some senior auditors showed doubt and resistance to RPA. But they were more open to RPA once they were shown its capabilities.
IT Governance	COE becoming a bottleneck	No	COE equivalent (IT department) did not become a bottleneck: 1) Small firm, and 2) Limited adoption of the bot.
	Auditability and Internal Control	Partially similar	Firm B has good management and control around the bot: 1) Internal control: credential control on who can use/manage/revise the bot; bots run on centralized virtual machine, and 2) Auditability: IT can provide a log of activities related to the bot.
Data Security and Privacy	Fraud and errors	Yes	Due to unexpected exceptions in audit processes, the bot stopped at certain point. But after including those variants, the bot started to work again.
System Sustainability	RPA sustainability	Yes	Upgrading systems where the bots operate is cautiously considered because that may render existing bots obsolete.
Measures of RPA Success	Measure of success	Yes	The firm has not realized significant reduction in efficiency. The bots were deployed in virtual machines so it does not matter how much time the bot takes to complete the assigned task, but auditors can use that time to do something else.
Other – Trust and Communication	Communication barriers affect bot development	Yes	Ineffective communication between employees and bot developers caused unnecessary re-configuration of some parts of the bot.

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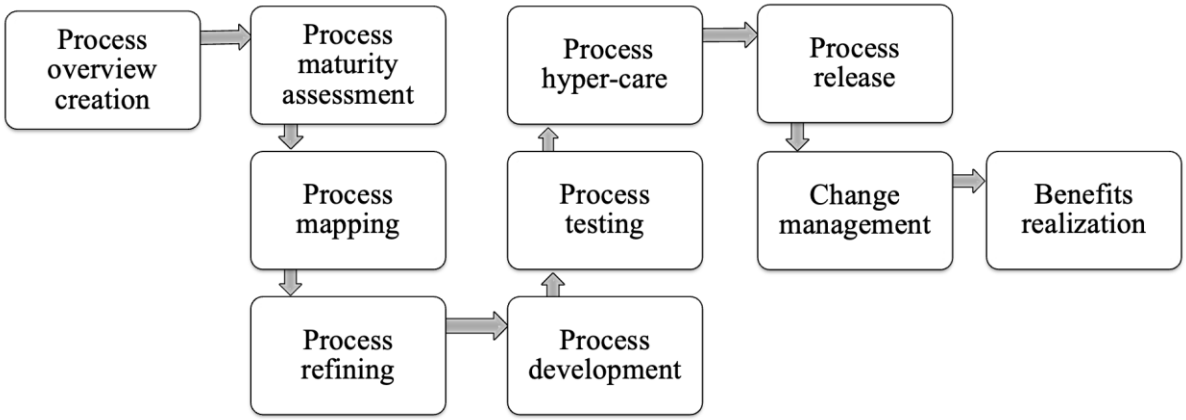
Figure 1. Firm A's RPA Implementation Approach



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APPENDIX

Figure A1. An Example of Process Overview

Area	**Sub-area**	**Process**
Operational accounting	Billing and collections	Create credit note for type x
		Create periodic bill for type y
		Create interim bill for type z
	Accounts receivable	…
General accounting	Allocations and adjust	…
	Journal entry processing	…

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Figure A2. An Example of Process Maturity Assessment Sheet

R	S	T	U	V	W
Process characteristics					
Is all input data in a digital format?	How structured is the input data?	Are there any calculation during the process?	How many different routes can the process take?	Output Type	Is the process documented?

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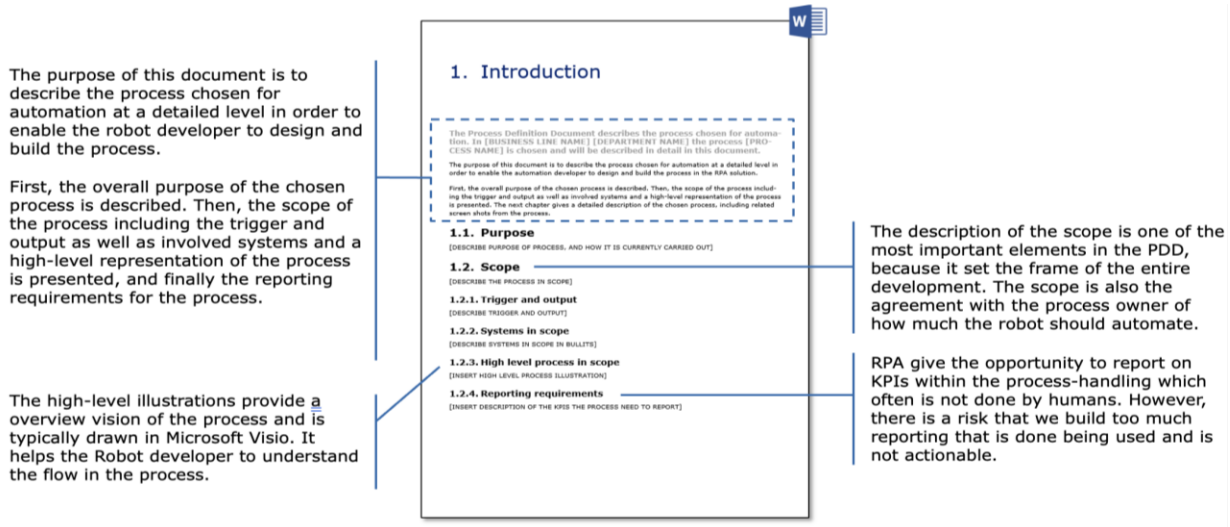
Figure A3. An Example of the Process Definition Document (PDD)



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